

Week 5 Report – Power BI Healthcare Dataset Project

Step 1: Data Preparation

- Imported the **Healthcare dataset** again to ensure clean data.
- Standardized elements, removed duplicates, and checked for errors.
- Created a new column **Age Category** using conditional logic:
- Senior Citizens → Age ≥ 60
- Adult → Age ≥ 49
- Grown Up → Age ≥ 29
- Young Adult → Age > 16
- Kid → else

👉 This categorizes patients into meaningful groups.

The screenshot shows a Power BI data table with the following columns: Name, Gender, Blood Type, Medical Condition, State of Admission, Doctor, and Hospital. The table contains 25 rows of patient data. The 'Reordered Columns' formula bar at the top shows the formula: `Table.ReorderColumns(Reordered Columns, {"Custom", "Age Category"})`. The 'Query Settings' pane on the right shows the 'Name' property set to 'Healthcare Dataset' and the 'Applied Steps' list.

Name	Gender	Blood Type	Medical Condition	State of Admission	Doctor	Hospital
1. JERRY JACKSON	Male	B-	Cancer	11-01-2024	MARTINE SMITH	St. Paul's
2. LUCAS TERRY	Male	A+	Obesity	22-09-2023	SAMANTHA DAVIES	St. Paul's
3. DANNY SMITH	Female	A-	Obesity	22-09-2022	TIGRARI MITCHELL	Cook P.C.
4. ANDREW WATTS	Female	O+	Diabetes	18-11-2020	KEVIN WELLS	Hennepin P.C.
5. ADRIENNE BELL	Female	AB+	Cancer	19-09-2022	KATHLEEN HANNA	White/White
6. EMILY JOHNSON	Male	A+	Asthma	20-12-2023	TAYLOR NEWTON	Waco-Hump
7. EDWARD EDWARDS	Female	AB-	Diabetes	07-12-2020	KELLY OLSON	Group Medical
8. CHRISTINA MARTINEZ	Female	A+	Cancer	28-12-2021	SUEANNE THOMAS	Powell-Robert
9. JEREMY AGUILAR	Male	AB+	Asthma	02-07-2020	DANIEL FERGUSON	St. Paul's
10. CHRISTOPHER BENE	Female	AB-	Cancer	23-05-2021	HEATHER DAY	Rede-Walker
11. RACHELLE DANIELS	Male	O+	Cancer	19-04-2020	JOHN DUNCAN	Islander-Ford
12. AMON MARTINEZ	Female	A-	Hypertension	13-09-2023	DOUGLAS MAYO	Lynn-Ryan
13. CONNOR HANSEN	Female	A+	Diabetes	17-12-2020	KENNETH FLETCHER	Powers-Miller
14. ROBERT BALER	Female	AB+	Asthma	23-05-2020	THERESA FLEMMING	Wyler-Schulz
15. BROCKE BERRY	Female	AB+	Cancer	08-10-2021	ROBERTA STEWART	Morris-Kelly
16. MRS. NATALIE GAMBLE	Female	AB-	Obesity	05-10-2023	MARIA DOUGHERTY	Chase-William
17. HARRY PERROWE	Female	A+	Asthma	27-06-2020	EMILY BUNCHER	Constance-W.
18. MRS. JANE CAMPBELL	Male	AB-	Obesity	08-03-2020	JUSTIN KRA	Fernex, and H.
19. LUKE BURGESS	Female	A+	Hypertension	04-03-2021	JUSTIN MOORE JR.	Islander-P.C.
20. DANIEL SCHWARTZ	Male	B+	Asthma	15-12-2022	DEREK GALLOWAY	Hammond-L.
21. TIMOTHY BURNS	Female	A-	Asthma	28-06-2023	KRISTA SMITH	James LLC
22. CHRISTOPHER BRADY	Male	B+	Asthma	11-03-2020	GREGORY SMITH	William-Dav
23. KATHLEEN STEWART	Female	O+	Asthma	17-03-2022	VANDRA NEWTON	Clark-Meyer
24. DR. OLIVER THOMPSON	Male	A+	Asthma	02-09-2021	DOMINIC MARTINEZ MD	and Sons Ltd
25.						

Step 3: Table Creation

1. Patients Table

- Columns: Age, Age Category, Blood Type, Gender, Name.
- Added PatientID (Index column starting at 1001).

2. Admission Table

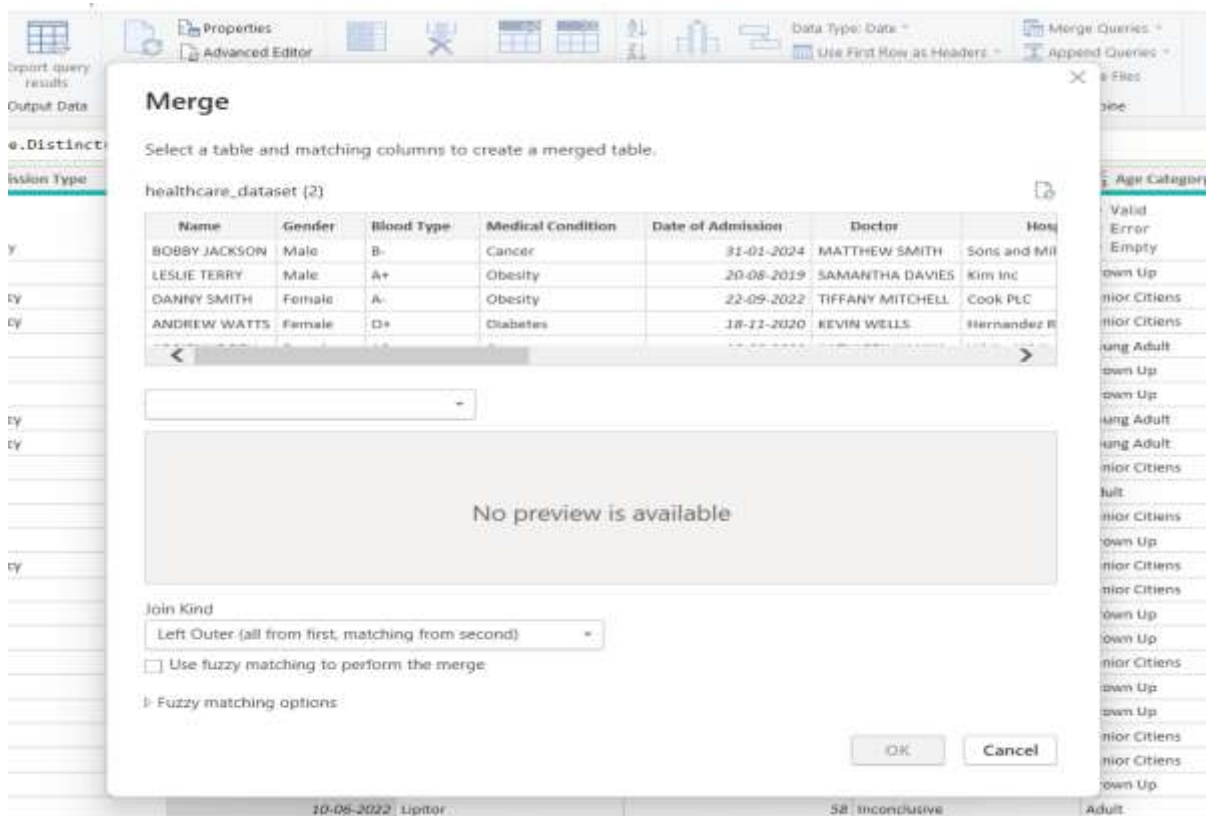
- Columns: Admission Type, AdmissionID, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results.

b. Added **PatientID** (Index column).

3. Merging Steps:

- Select transform data on your home screen
- Duplicate the original dataset
- Remove Duplicates
- Select merge Queries from the menu bar
- Your Duplicated table will be appearing on the top and a drop down box will be seen in the pop up window.
- Select it and first choose the Admission table.
- Touch and Select all the columns in the admission table except the Admission id and select only the same set of columns in the duplicated table appearing above. And click ok and finish it.
- Now the Admission Table will be created in the new column of the duplicate table, Select only the Admission id from that.

Repeat the same for Patient table also so both the Patient id and Admission id will be merged to this table .



O+	Arthritis	12-05-2022	VANESSA NEWTON
A+	Asthma	02-08-2021	DONNA MARTINEZ MD
A+	Arthritis	05-05-2022	DEBRA LEWIS MD MSN

- **Billing Table**

- Rename the Duplicate Table as Billing Table and keep only these columns and Delete the rest.
- Columns: Billing Amount, Date of Admission, Discharge Date, Insurance Provider.
- Merged with Patients (PatientID) and Admissions (AdmissionID).

👉 Now you have three logical tables ready for modeling.

Step 4: Relationship Generation

- Go to **Model View**.
- Relationships were automatically created:
- Verified the **Star Schema** structure for clean analysis.

Properties >>

Relationship

Table

Billing Table

Column

Admission Ta...

Cardinality

Many to one (*:1)

Table

Admission Ta...

Column

Admisssion ID

Make this relationship active

Yes

Cross-filter direction

Single

Apply security filter in both directions

No

Apply changes

Open relationship editor

Properties >>

Relationship

Table

Billing Table

Column

Patient Table....

Cardinality

Many to one (*:1)

Table

Patient Table

Column

Patient ID

Make this relationship active

Yes

Cross-filter direction

Single

Apply security filter in both directions

No

Apply changes

Open relationship editor

